Chiara Tramarin

MSc in Physics of Data
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ChiaTrama
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Education

•Master's Degree in Physics of Data — Mar. 2024 – Oct. 2026
University of Padua, Padua, Italy — Grade: 110/110

•Bachelor's Degree in Physics — Oct. 2018 – Apr. 2024
University of Padua, Padua, Italy — Grade: 92/110
Erasmus exchange: Universidad Autónoma de Madrid, Spain · Sept. 2022 – Mar. 2023

Research Experience

•Research Intern – IFIMAC, Madrid — Mar. – Sept. 2026

Conducted computational research on DNA mechanics and nucleosome stability using coarse-grained molecular dynamics simulations. Performed free-energy calculations through thermodynamic integration and analyzed simulation data using Python to investigate sequence-dependent physical properties. Ran simulations on an HPC GPU cluster using Linux, Bash and SLURM.

Master's thesis: *Investigating sequence-dependent DNA mechanics and nucleosome stability through coarse-grained simulations*

•Bachelor's Thesis Research – University of Padua — Nov. 2023 – Apr. 2024

Performed experimental measurements on a lithium-niobate ($LiNbO_3$) opto-microfluidic Lab-on-a-Chip for detecting transparent microplastics in water droplets, combining sample preparation, optical and electronic instrumentation (lasers, oscilloscopes and microfluidic control), and Python-based data processing and visualization.

Bachelor's thesis: *Optical sensing study of microplastics in droplets via refractive index contrast*

Related publication — Optics Express (2026)

Optofluidic light-droplet interaction for rapidly assessing the presence of plastic microspheres within aqueous suspensions

Selected Academic Projects

•N-Body Simulation on CPUs and GPUs
Developed a C++ gravitational N-body simulator with serial, OpenMP/SIMD and CUDA implementations. Validated energy conservation and benchmarked CPU/GPU performance across particle counts, OpenMP thread counts and CUDA block sizes.

•Bayesian Optimization
Implemented Gaussian process-based Bayesian optimization to optimize synthetic benchmark functions and tune CNN hyperparameters (learning rate, dropout and weight decay). Compared four acquisition strategies (EI, PI, UCB and Knowledge Gradient) against random search.

•Simulation-Based Inference for a 2D Random Walk
Applied simulation-based inference with neural posterior estimation to recover parameters of a stochastic 2D random walk, validated on an analytically solvable quartic model, and extended to a red-blood-cell model not accessible analytically. Optimized simulation performance with Numba to achieve C-like speed.

•Car Image Classification with CNNs
Developed PyTorch models for fine-grained vehicle classification and image verification on CompCars. Applied transfer learning with pretrained CNNs and Siamese networks, comparing architectures and performance.

•Distributed K-Means with Dask
Implemented and benchmarked a distributed K-Means‖ algorithm with Dask, extending the dask-ml implementation with weighted centroid updates. Configured VM clusters via SSH, optimized memory and thread allocation, and analyzed scalability on large datasets (KDD Cup 1999).

Technical Skills & Languages

Programming Languages Python, C++, R, MATLAB, SQL
High-Performance & Parallel Computing CUDA, OpenMP, Dask, Numba, Slurm
Software & Tools Git, Docker, Linux, Jupyter, VS Code, LaTeX, Microsoft Office (Excel, Word, PowerPoint)
Languages Italian (native), English (advanced), Spanish (advanced)